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LIGHT EMITTING DEVICE, WEARABLE DEVICE, DISPLAY DEVICE, PHOTOELECTRIC CONVERSION DEVICE, ELECTRONIC APPARATUS, ILLUMINATION DEVICE, AND MOVING BODY

Abstract

Alight emitting device including, on a main surface of a semiconductor substrate, a display region in which pixels are arranged, is provided. The main surface includes a first region in which a first conductive pattern is arranged, and a second region which is adjacent to the first region and in which a second conductive pattern is arranged. Each of the pixels includes an organic light emitting element and a driving transistor. Each of the first conductive pattern and the second conductive pattern includes a conductive pattern connected to a gate electrode of the driving transistor in one of the pixels. The first region and the second region are adjacent at a boundary with a deviation between the first conductive pattern and the second conductive pattern, and the boundary does not pass through a center of the display region.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a light emitting device, a wearable device, a display device, a photoelectric conversion device, an electronic apparatus, an illumination device, and a moving body.

Description of the Related Art

[0002] Recently, in a light emitting device using an organic light emitting element such as an organic light emitting diode (OLED), a light emitting device having a higher resolution has been demanded along with the development of the virtual reality (VR) and augmented reality (AR) technologies. To increase the resolution, there is known a micro OLED in which transistors are formed on a silicon wafer using a semiconductor fine-process technology. In addition, a light emitting device including more pixels is required, and as a result, a light emitting device having a larger size is demanded. However, a size that an exposure apparatus used in the semiconductor fine-process technology can expose is generally about 33 mm×26 mm. When manufacturing a light emitting device having a large size, it is considered that a region to expose is divided into two or more regions, and stitching exposure is performed. Japanese Patent Laid-Open No. 2021-064001 shows flexibly setting the position of a stitching portion between adjacent patterns in a semiconductor device.

SUMMARY OF THE INVENTION

[0003] Japanese Patent Laid-Open No. 2021-064001 shows setting the stitching portion to a visually unnoticeable position in an image capturing device or the like. However, an influence on image quality derived from the stitching exposure in a light emitting device using an organic light emitting element is not disclosed in Japanese Patent Laid-Open No. 2021-064001.

[0004] Some embodiments of the present disclosure provide a technique advantageous in suppressing image quality degradation caused by stitching exposure in a light emitting device using an organic light emitting element.

[0005] According to some embodiments, a light emitting device comprising, on a main surface of a semiconductor substrate, a display region in which a plurality of pixels are arranged, wherein the main surface includes a first region in which a first conductive pattern is arranged, and a second region which is adjacent to the first region and in which a second conductive pattern is arranged, each of the plurality of pixels includes an organic light emitting element, and a driving transistor configured to supply a current according to a luminance signal to the organic light emitting element, each of the first conductive pattern and the second conductive pattern includes a conductive pattern connected to a gate electrode of the driving transistor in one of the plurality of pixels, the first region and the second region are adjacent at a boundary with a deviation between

the first conductive pattern and the second conductive pattern, and the boundary does not pass through a center of the display region, is provided.

[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a sectional view showing an example of the configuration of a light emitting device according to the embodiment.

[0008] FIG. 2 is a circuit diagram showing an example of the configuration of a pixel of the light emitting device shown in FIG. 1.

[0009] FIG. 3 is a plan view showing an example of the arrangement of the pixels shown in FIG. 2.

[0010] FIG. 4 is a plan view showing an example of the configuration of the light emitting device shown in FIG. 1.

[0011] FIG. 5 is a plan view showing an example of the configuration of the light emitting device shown in FIG. 1.

[0012] FIG. 6 is a plan view showing an example of the configuration of the light emitting device shown in FIG. 1.

[0013] FIG. 7 is a plan view showing an example of the configuration of the light emitting device shown in FIG. 1.

[0014] FIGS. 8A and 8B are views showing an example of the configuration of a head mounted display using the light emitting device shown in FIG. 1.

[0015] FIGS. 9A and 9B are sectional views showing an example of the configuration of the pixel of the light emitting device shown in FIG. 1.

[0016] FIG. 10 is a view showing an example of a display device using the light emitting device according to the embodiment.

[0017] FIG. 11 is a view showing an example of a photoelectric conversion device using the light emitting device according to the embodiment.

[0018] FIG. 12 is a view showing an example of an electronic apparatus using the light emitting device according to the embodiment.

[0019] FIGS. 13A and 13B are views showing an example of a display device using the light emitting device according to the embodiment.

[0020] FIG. 14 is a view showing an example of an illumination device using the light emitting device according to the embodiment.

[0021] FIG. 15 is a view showing an example of a moving body using the light emitting device according to the embodiment.

[0022] FIGS. 16A and 16B are views showing an example of a wearable device using the light emitting device according to the embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0024] A light emitting device according to an embodiment of the present disclosure will be described with reference to FIGS. 1 to 8A and 8B. FIG. 1 is a sectional view showing an example of the configuration of a light emitting device 100 according to this embodiment. FIG. 2 is a view

showing an example of the configuration of a driving circuit of a pixel PIX arranged in the light emitting device **100**. FIG. 3 is a plan view showing an example of the arrangement of the driving circuit shown in FIG. 2. FIG. 4 is a plan view showing an example of the configuration of the light emitting device **100**.

[0025] The light emitting device **100** includes, on a main surface **151** of a substrate **115**, a display region **402** (also called an effective pixel region) in which a plurality of pixels PIX are arranged. For the substrate **115**, a semiconductor such as silicon can be used. A wiring structure **101** is arranged on the substrate **115** provided with a transistor TR. The wiring structure **101** includes an insulator using silicon oxide or the like, wiring patterns arranged in the insulator, plugs for connecting the wiring patterns to each other, and the like. A lower electrode **110** that forms an organic light emitting element, an insulating layer **111**, an organic layer **112** including a light emitting layer, and an upper electrode **113** are arranged on the wiring structure **101**. A protection layer **114** can be arranged on the organic light emitting element.

[0026] The transistor TR includes, for example, a gate electrode **102**. FIG. 1 shows the wiring structure **101** having a two-layered structure including a wiring layer in which a wiring pattern **104** is arranged, and a wiring layer in which a wiring pattern **106** is arranged. Plugs **103** connected to the gate electrode **102** or the substrate **115** are arranged in the wiring structure **101**. The plugs **103** connect the gate electrode **102** and the wiring pattern **104**, and the substrate **115** and the wiring pattern **104**. Also, a plug **105** that connects the wiring pattern **104** and the wiring pattern **106** and a plug **107** that connects the wiring pattern **106** and the lower electrode **110** are arranged in the wiring structure **101**. In the configuration shown in FIG. 1, there exist two wiring layers each having a wiring pattern arranged therein. However, there may be one wiring layer or three or more wiring layers in accordance with the configuration of the light emitting device **100**.

[0027] FIG. 2 shows an example of the configuration of the circuit of the pixel PIX according to this embodiment. The pixel PIX can include an organic light emitting element **201**, a driving transistor **202**, and a write transistor **203**. The above-described transistor TR corresponds to the driving transistor **202** or the write transistor **203**. In the write transistor **203**, the gate electrode **102** is connected to a signal line SEL, and one main terminal (a source in the configuration shown in FIG. 2) is connected to a signal line DATA. The gate electrode **102** of the driving transistor **202** is connected to the other main terminal (a drain in the configuration shown in FIG. 2) of the write transistor **203**. One main terminal (a source in the configuration shown in FIG. 2) of the driving transistor **202** is connected to a power supply line VDD, and the other main terminal (a drain in the configuration shown in FIG. 2) of the driving transistor **202** is connected to one main terminal of the organic light emitting element **201**. The other main terminal of the organic light emitting element **201** is connected to a power supply line VSS. If the write transistor **203** is rendered conductive in accordance with a signal supplied to the signal line SEL, a predetermined potential is supplied as a luminance signal from the signal line DATA to the gate electrode **102** of the driving transistor **202**. The driving transistor **202** supplies a current according to the potential of the luminance signal to the organic light emitting element **201**. The organic light emitting element **201** thus emits light in a luminance according to the luminance signal (current driving).

[0028] FIG. 3 is a plan view showing an example of the arrangement of two pixels PIXa and PIXb according to this embodiment. In the actual pixels PIXa and PIXb, components that are not shown in FIG. 3 exist, and FIG. 3 shows only minimum components necessary for the description. As shown in FIG. 2, the pixels PIXa and PIXb include driving transistors **202a** and **202b** and write transistors **203a** and **203b**, respectively. The driving transistors **202a** and **202b** include source/drain regions **116a** and **116c** and gate electrodes **102a** and **102c** provided on the substrate **115**, respectively. The write transistors **203a** and **203b** include source/drain regions **116b** and **116d** and gate electrodes **102b** and **102d** provided on the substrate **115**, respectively.

[0029] The gate electrode **102a** of the driving transistor **202a** and the drain of the write transistor **203a** are connected via plugs **103a** and **103b**, a wiring pattern **104a**, and a plug **103c**. The wiring

pattern **104a** forms a part of the wiring pattern **104** shown in FIG. 1. Also, the wiring pattern **104a** may be connected, via plugs **105a** and **105b**, to a wiring pattern **106a** that forms a part of the wiring pattern **106** shown in FIG. 1. As shown in FIG. 3, a plug **107a** may be connected to the wiring pattern **106a**. The wiring patterns **104a** and **106a**, and the plugs **103a**, **103b**, **103c**, **105a**, **105b**, and **107a** are patterns connected to the gate electrode **102a** of the driving transistor **202a**. It can be said that the wiring patterns **104a** and **106a**, and the plugs **103a**, **103b**, **103c**, **105a**, **105b**, and **107a** are patterns that are equipotential to the gate electrode **102a** of the driving transistor **202a** if wiring resistances and a transient potential variation are neglected. A wiring pattern **104b** forms a part of the wiring pattern **104** shown in FIG. 1. Also, the wiring pattern **104b** is a pattern connected to the source of the write transistor **203a** via a plug **103d**. A wiring pattern **104c** forms a part of the wiring pattern **104** shown in FIG. 1. The wiring pattern **104c** is a pattern connected to the gate electrode **102b** of the write transistor **203a** via a plug **103e**.

[0030] Similarly, the gate electrode **102c** of the driving transistor **202b** and the drain of the write transistor **203a** are connected via plugs **103f** and **103g**, a wiring pattern **104d**, and a plug **103h**. The wiring pattern **104d** forms a part of the wiring pattern **104** shown in FIG. 1. Also, the wiring pattern **104d** may be connected, via plugs **105c** and **105d**, to a wiring pattern **106b** that forms a part of the wiring pattern **106** shown in FIG. 1. As shown in FIG. 3, a plug **107b** may be connected to the wiring pattern **106b**. The wiring patterns **104d** and **106b**, and the plugs **103f**, **103g**, **103h**, **105c**, **105d**, and **107b** are patterns connected to the gate electrode **102c** of the driving transistor **202b**. It can be said that the wiring patterns **104d** and **106b**, and the plugs **103f**, **103g**, **103h**, **105c**, **105d**, and **107b** are patterns that are equipotential to the gate electrode **102a** of the driving transistor **202a** if wiring resistances and a transient potential variation are neglected. The wiring pattern **104e** forms a part of the wiring pattern **104** shown in FIG. 1. Also, the wiring pattern **104e** is a pattern connected to the source of the write transistor **203b** via a plug **103i**. A wiring pattern **104f** forms a part of the wiring pattern **104** shown in FIG. 1. The wiring pattern **104f** is a pattern connected to the gate electrode **102d** of the write transistor **203b** via a plug **103j**.

[0031] In this embodiment, when manufacturing the light emitting device **100**, for some components, the region to expose is divided into two or more regions, and stitching exposure is performed. For example, in an exposure process when forming the wiring pattern **104** shown in FIG. 3, stitching exposure is executed using, as a boundary, a boundary **300** between the pixel **PIXa** and the pixel **PIXb**. The wiring pattern **104** is a pattern arranged in the wiring layer closest to the gate electrode **102** in a plurality of wiring layers arranged in the display region **402**. On the other hand, the gate electrode **102**, the plug **103**, the plug **105**, the wiring pattern **106**, and the plug **107**, and the like may be formed by one-shot exposure. However, the pattern formed using stitching exposure is not limited to the wiring pattern **104**. For example, the gate electrode **102** may be formed using stitching exposure. Also, for example, the wiring pattern **106** that is a pattern arranged in the second closest wiring layer to the gate electrode **102** in the plurality of wiring layers arranged in the display region **402** may be formed using stitching exposure. Furthermore, for example, one of the plug **103** connected to the gate electrode **102** or the substrate **115**, the plug **105** connected to the wiring pattern **104**, and the plug **107** connected to the wiring pattern **106** may be formed using stitching exposure.

[0032] For example, in a layer where fine pattern formation is needed, stitching exposure by ArF exposure may be used, and in a layer where a relatively large pattern is formed, one-shot exposure by KrF exposure or i-line exposure may be used. Wiring patterns and plugs formed using stitching exposure will sometimes be referred to as conductive patterns **121** hereinafter. In the configuration shown in FIG. 3, the wiring pattern **104** (the wiring patterns **104a** to **104f**) is the conductive pattern **121** formed using stitching exposure.

[0033] If stitching exposure is used, the line width of the wiring pattern **104** that is the conductive pattern **121** formed using stitching exposure may have a different value for each divided and exposed region. In addition, the overlay amount of the wiring pattern **104** to the patterns in another

layer (for example, a layer formed using one-shot exposure), for example, the gate electrode **102**, the plugs **103** and **105**, the wiring pattern **106**, and the like may also change. More specifically, the line width may change at the boundary **300** between the wiring pattern **104a** included in a conductive pattern **121a** and the wiring pattern **104d** included in a conductive pattern **121b**. Similarly, the line width may change between the wiring pattern **104c** (conductive pattern **121a**) and the wiring pattern **104f** (conductive pattern **121b**). Accordingly, the parasitic capacitance between the wiring pattern **104a** and the wiring pattern **104c** and the parasitic capacitance between the wiring pattern **104d** and the wiring pattern **104f** may have different values. The wiring patterns **104c** and **104f** are the signal lines SEL shown in FIG. 2. That is, the parasitic capacitance between the signal line SEL and the gate electrode **102** of the driving transistor **202** can vary at the boundary **300** of stitching exposure.

[0034] The parasitic capacitance between the signal line SEL and the gate electrode **102** of the driving transistor **202** affects a voltage applied from the signal line DATA to the gate electrode **102** of the driving transistor **202** when the write transistor **203** is rendered conductive. If the value of the parasitic capacitance is different, the voltage applied to the gate electrode **102** of the driving transistor **202** changes even in a case where the supplied luminance signal is the same, and the amount of the current flowing to the organic light emitting element **201** changes. That is, the amount of the current flowing to the organic light emitting element **201** varies for each exposure region in stitching exposure, and the light emission intensity of the organic light emitting element **201** may change at the boundary **300** of stitching exposure.

[0035] As shown in FIG. 4, the display region **402** including the above-described pixels PIX is arranged on the main surface **151** of the substrate **115**. A dummy pixel region **400** in which dummy pixels are arranged is arranged around the display region **402**. A dummy pixel can be a pixel that has the same configuration as the pixel PIX but does not emit light. For example, in the dummy pixel, the plug **107** that connects the wiring pattern **106** and the lower electrode **110** may not be arranged. A peripheral region **401** is arranged around the dummy pixel region **400**. A circuit configured to operate the pixel PIX, and the like can be arranged in the peripheral region **401**.

[0036] On the main surface **151** of the substrate **115**, the conductive pattern **121** (for example, the above-described wiring pattern **104**) formed by stitching exposure at the boundary **300**, as described above, is arranged. It can be said that the main surface **151** of the substrate **115** includes, on both sides of the boundary **300**, a region **411a** where the conductive pattern **121a** (for example, the wiring patterns **104a** to **104c**) formed using stitching exposure is arranged and a region **411b** where the conductive pattern **121b** (for example, the wiring patterns **104d** to **104f**) is arranged. In this case, as described above, for example, the conductive pattern **121a** arranged in the region **411a** and the conductive pattern **121b** arranged in the region **411b** may have different line widths. In addition, the relative position with respect to the layer formed using one-shot exposure may be different between the conductive pattern **121a** arranged in the region **411a** and the conductive pattern **121b** arranged in the region **411b**. Hence, the region **411a** and the region **411b** are adjacent to each other at the boundary with the deviation between the conductive pattern **121a** and the conductive pattern **121b**.

[0037] The display region **402** can have, for example, a substantially rectangular shape having sides **412a** to **412d**, as shown in FIG. 4. The sides **412a** to **412d** can be virtual lines that connect the centers of the pixels PIX arranged at the vertices of the display region **402**. Here, the center of the pixel PIX can be defined as the position of the geometrical center of gravity in the whole configuration or a specific component of the pixel PIX in the orthogonal projection to the main surface **151** of the substrate **115**. For example, the center of the pixel PIX may be the position of the geometrical center of gravity of a region of the pixel PIX, where light is emitted, in the orthogonal projection to the main surface **151** of the substrate **115**. The position of the geometrical center of gravity of the rectangle defined by the sides **412a** to **412d** in the orthogonal projection to the main surface **151** of the substrate **115** is defined as a center **410** of the display region. In this

case, the boundary **300** when executing stitching exposure is arranged not to pass through the center of the display region **402**.

[0038] As described above, in the conductive pattern **121** formed using stitching exposure, a pattern size difference or misalignment occurs between regions **411**. This may change the light emission intensity of the organic light emitting element **201** at the boundary **300** of stitching exposure. This can prevent the boundary **300** of stitching exposure from being arranged at the center of the display region **402** which is most likely to stand out to a viewer. As a result, even if the boundary **300** passes through the display region **402**, and the light emission intensity changes between the region **411a** and the region **411b** at the boundary **300**, it is possible to suppress visually recognizing the difference of the light emission intensity. As a result, in the light emitting device **100** using the organic light emitting element **201**, image quality degradation caused by stitching exposure is suppressed.

[0039] For example, as shown in FIG. **4**, the boundary **300** may be arranged between the center **410** of the display region **402** and the side **412a** that forms the outer edge of the display region **402**. In this case, the boundary **300** may extend parallel to the direction in which the side **412a** extends. In this case, it can be said that the boundary **300** is arranged between a center line **403** that passes through the center **410** of the display region **402** and is parallel to the direction in which the side **412a** extends and the side **412a** that forms the outer edge of the display region **402**. The configuration shown in FIG. **4** shows an example in which, in the orthogonal projection to the main surface **151** of the substrate **115**, the rectangular shape of the outer edge of the display region **402** has long sides and short sides, and the boundary **300** is arranged parallel to the side **412a** that forms a short side. However, the present invention is not limited to this, and the boundary **300** may be arranged parallel to the sides **412b** and **412d** that form the long sides. Also, the outer edge of the display region **402** may have a square shape.

[0040] An example in which the boundary **300** of stitching exposure is arranged at a position more difficult for a person to visually recognize will be described next with reference to FIG. **5**. The relationship of the components in the orthogonal projection to the main surface **151** of the substrate **115** will be shown below unless it is specifically stated otherwise in this specification.

[0041] As described above, the boundary **300** of stitching exposure does not pass through the center of the display region **402**. Also, in the configuration shown in FIG. **5**, in a direction crossing the direction in which the boundary **300** extends, the boundary **300** is arranged between the side **412a** among sides **412** that form the outer edge of the display region **402** and the center **410** of the display region **402**. At this time, a length **502** between the boundary **300** and the center **410** of the display region **402** may be equal to or more than a length **501** between the boundary **300** and the side **412a**. As shown in FIG. **5**, the length between the side **412a** and the side **412c**, which are parallel to each other and form the outer edge of the display region **402**, is defined as a length **511**. The length **511** is the length of the display region **402** in the direction crossing the boundary **300**. In this case, the boundary **300** is arranged within the range of a distance $\frac{1}{4}$ the length **511** from the side **412a**. Thus, the boundary **300** is apart from the center of the display region **402**, and visually recognizing the boundary **300** is suppressed.

[0042] Also, for example, the outer edge of the substrate **115** includes a portion **500**, and the boundary **300** is arranged between the portion **500** of the substrate **115** and the center **410** of the display region **402** in the direction crossing the direction in which the boundary **300** extends. At this time, the length **502** between the boundary **300** and the center **410** of the display region **402** may be equal to or more than a length **503** between the boundary **300** and the portion **500** of the substrate **115**.

[0043] Also, assume that the boundary **300** is arranged between the side **412a** and the center **410** of the display region **402**, as shown in FIG. **5**, and the visual field angle of the display region **402** in the light emitting device **100** is A degree. The visual field angle of the display region **402** is the design value of the visual field when the user views the display region **402** through an eyepiece

optical system (for example, an optical system arranged in a viewfinder **1101** shown in FIG. **11**) arranged in the light emitting device **100**. In this case, the boundary **300** of stitching exposure may be arranged within the range of $\{(A/2-30)/A\} \times 100[\%]$ from the side **412a** with respect to a length **511** of the display region in the direction crossing the boundary **300**. For example, the visual field angle can be set to 105° to 120°. This is because the visual field to be clearly recognized by person's eyes in the horizontal direction is about $\pm 30^\circ$. Hence, the boundary **300** may be arranged, for example, within the range of 30° or more in the direction crossing the boundary **300** from the center of the visual field angle through the eyepiece optical system. When the position of the boundary **300** of stitching exposure is deviated from the range of the visual field that a person readily recognizes, the difference of the light emission intensity caused by stitching exposure becomes more difficult to recognize by human vision.

[0044] The above description has been made using an example in which the boundary **300** is arranged between the center **410** of the display region **402** and the side **412a** that is a short side among the sides **412** that form the outer edge of the display region **402**. However, the present invention is not limited to this, and the same applies to a case where the boundary **300** is arranged between the center **410** of the display region **402** and the side **412b** or **412d** that is a long side among the sides **412** that form the outer edge of the display region **402**. The short sides (sides **412a** and **412c**) and the long sides (sides **412b** and **412d**) may be replaced with each other below unless it is specifically explicitly stated.

[0045] There exists a technique called foveated rendering that lowers the resolution of the display region **402** generally in a peripheral visual field that a person is not gazing, thereby reducing a load in rendering processing. The present disclosure can obtain a higher effect by arranging the boundary **300** of stitching exposure in the region with the lower resolution when executing the foveated rendering.

[0046] FIG. **6** is a view showing the display region **402** in a case where foveated rendering is performed. The display region **402** includes a display region **601**, and a display region **602** in which the resolution of a displayed image is lower than that in the display region **601**. The display region **601** is arranged at the center portion of the display region **402**. In this case, the boundary **300** of stitching exposure passes through the display region **602**. The boundary **300** may not pass through the display region **601**. This makes it difficult for human vision to recognize the difference of the light emission intensity caused by stitching exposure.

[0047] For example, assume that of the display region **402**, a region at the center whose size is $\frac{1}{2}$ in the short side direction and the long side direction is set as the display region **601**. In this case, the boundary **300** is arranged within the range of a distance $\frac{1}{4}$ the length **511** from the side **412a**. The effect of the present disclosure can thus be obtained. Here, the display region **601** may be, for example, a region in which the arrangement density of the pixels PIX is higher than in the display region **602**. Also, for example, the display region **602** may be a region in which the resolution of an image to be displayed is made lower than that in the display region **601** by signal processing.

[0048] In addition, the conductive pattern **121** formed using stitching exposure can be arranged even in a peripheral region **401**. Hence, as shown in FIG. **7**, the boundary **300** of stitching exposure may be arranged in the peripheral region **401**. That is, the boundary **300** may not pass through the display region **402**. In this case, the difference of the light emission intensity in the display region **402** caused by arranging the boundary **300** in the display region **402** hardly occurs. As a result, it is possible to further suppress image quality degradation of a displayed image caused by the boundary **300** of stitching exposure.

[0049] Next, an example in which the above-described light emitting device **100** is applied to a wearable device such as a head mounted display or smartglasses will be described. Each of FIGS. **8A** and **8B** shows a display device **800** included in a wearable device and configured to display an image. The display device **800** includes a plurality of light emitting devices including a light emitting device **100a** arranged in correspondence with the left eye when a user wears it, and a light

emitting device **100b** arranged in correspondence with the right eye. In this case, as shown in FIG. **8A**, the direction in which a boundary **300a** of the light emitting device **100a** is arranged with respect to a center **410a** of the display region **402** of the light emitting device **100a** when the user wears the wearable device and the direction in which a boundary **300b** of the light emitting device **100b** is arranged with respect to a center **410b** of the display region **402** of the light emitting device **100b** may be different from each other. More specifically, the boundary **300a** of the light emitting device **100a** may be arranged on the left side with respect to the center **410a** of the display region **402** of the light emitting device **100a**, and the boundary **300b** of the light emitting device **100b** may be arranged on the right side with respect to the center **410b** of the display region **402** of the light emitting device **100b**. Thus, in the light emitting devices **100a** and **100b** arranged in correspondence with the left and right eyes, respectively, the boundaries **300a** and **300b** of stitching exposure are arranged outside the visual field. It is therefore possible to suppress visually recognizing the difference of the light emission intensity caused by the boundary **300** of stitching exposure. As a result, in the wearable device including the display device **800** including the light emitting devices **100a** and **100b**, image quality degradation caused by stitching exposure in the light emitting devices **100a** and **100b** is suppressed.

[0050] Also, for example, as shown in FIG. **8B**, when the user wears the wearable device, the boundaries **300a** and **300b** of the light emitting devices **100a** and **100b** may be arranged on the upper side with respect to the centers **410a** and **410b** of the display regions **402** of the light emitting devices **100a** and **100b**. The visual field to be clearly recognized by person's eyes in the vertical direction is about 15° on the upper side and about 20° on the lower side. Also, when relaxed, people are known to direct the line of sight downward. For this reason, the boundary **300** of stitching exposure is set on the upper side of the centers **410a** and **410b** of the display regions **402** in the light emitting devices **100**. This makes it possible to suppress visually recognizing the difference of the light emission intensity caused by the boundary **300** of stitching exposure.

[0051] Here, application examples in which the light emitting device **100** according to this embodiment is applied to a display device, a photoelectric conversion device, an electronic apparatus, an illumination device, a moving body, and a wearable device will be described with reference to FIGS. **9A** and **9B** to **16A** and **16B**. The description will be given assuming that, for example, an organic light emitting element (OLED) such as an organic EL element using an organic light emitting material is arranged in the pixel PIX of the light emitting device **100**. Details of each component arranged in the pixel PIX of the light emitting device **100** described above will be described first, and the application examples will be described after that.

[0052] The organic light emitting element according to an embodiment of the present disclosure includes a first electrode, a second electrode, and an organic compound layer arranged between these electrodes. One of the first electrode and the second electrode is an anode, and the other is a cathode. In the organic light emitting element according to this embodiment, the organic compound layer may be either a single layer or a stacked body formed by a plurality of layers as long as it includes a light emitting layer. Here, if the organic compound layer is a stacked body formed from a plurality of layers, the organic compound layer may include a hole injection layer, a hole transport layer, an electron blocking layer, a hole/exciton blocking layer, an electron transport layer, an electron injection layer, and the like in addition to the light emitting layer. The light emitting layer may be a single layer or a stacked body formed from a plurality of layers. If the light emitting layer includes a plurality of layers, a charge generation layer may be arranged between the light emitting layers. The charge generation layer may be made of a compound having the LUMO lower than that of the hole transport layer, and the LUMO of the charge generation layer may be lower than the HOMO of the hole transport layer. Here, the molecular orbital energy of the organic compound layer may be the molecular orbital energy of the organic compound with the largest weight ratio in the organic compound layer.

[0053] The description is given here assuming that the closer the HOMO and LUMO are to the

vacuum level, the “higher” they are. When the LUMO of the charge generation layer is lower than the HOMO of the hole transport layer, the LUMO of the charge generation layer is closer to the vacuum level than the HOMO of the hole transport layer.

[0054] The HOMO and LUMO in this specification can be calculated using molecular orbital calculation. The molecular orbital calculation is executed by a Density Functional Theory (DFT) or the like. A functional may be calculated using B3LYP, and a basic function may be calculated using 6-31G*. Note that molecular orbital calculation can be executed using, for example, Gaussian 09 (Gaussian 09, Revision C.01, M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V.N. Staroverov, T. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, O. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, and D. J. Fox, Gaussian, Inc., Wallingford CT, 2010.)

[0055] The HOMO and LUMO in this specification can be calculated using the ionization potential and band gap. The HOMO can be estimated by measuring the ionization potential. The ionization potential can be measured by dissolving the compound to be measured in a solvent such as toluene and using a measuring device such as AC-3. The band gap can be measured by dissolving the compound to be measured in a solvent such as toluene and irradiating it with excitation light. The band gap can be measured by measuring the absorption edge of the excitation light. Alternatively, the band gap can be measured by depositing the compound to be measured on a substrate such as glass, and exposing the deposited film to excitation light. The band gap can be measured by measuring the absorption edge of the absorption spectrum at which the deposited film absorbs excitation light.

[0056] The LUMO can be calculated using the band gap and ionization potential value. The LUMO can be estimated by subtracting the ionization potential value from the band gap.

[0057] The LUMO can also be estimated from the reduction potential. For example, the one-electron reduction potential is estimated using cyclic voltammetry (CV) measurement. The CV measurement can be performed, for example, in a DMF solution of 0.1 M tetrabutylammonium perchlorate using a reference electrode of Ag/Ag^{sup.}+, a counter electrode of Pt, and a working electrode of glassy carbon. The LUMO can be estimated by adding -4.8 eV to the difference between the reduction potential of the obtained compound and that of ferrocene.

[0058] A conventionally known low molecular and high molecular hole injection compound or hole transport compound, a compound serving as a host, a light emitting compound, an electron injection compound or electron transport compound, or the like can be used together as needed. Examples of these compounds will be described below.

[0059] As a hole injection/transport material, a material that has a high hole mobility such that hole injection from the anode is facilitated, and injected holes can be transported to the light emitting layer can suitably be used. Also, a material having a high glass transition point temperature can suitably be used to reduce degradation of film quality such as crystallization in the organic light emitting element. Examples of low molecular and high molecular materials having hole injection/transport performance are a triarylamine derivative, an arylcarbazole derivative, a phenylenediamine derivative, a stilbene derivative, a phthalocyanine derivative, a porphyrin derivative, a poly(vinyl carbazole), a poly(thiophene), and other conductive polymers. The above-described hole injection/transport material can suitably be used for the electron blocking layer as well. Detailed examples of compounds used as the hole injection/transport material will be shown

below. The material is not limited to these.

##STR00001## ##STR00002## ##STR00003##

[0060] In the hole transport materials, HT16 to HT18 can decrease the driving voltage when used in a layer in contact with the anode. HT16 is widely used in an organic light emitting element. HT2, HT3, HT4, HT5, HT6, HT10, and HT12 can be used in an organic compound layer adjacent to HT16. A plurality of materials may be used in one organic compound layer.

[0061] Examples of the light emitting material mainly concerning the light emitting function are condensed-ring compounds (for example, a fluorene derivative, a naphthalene derivative, a pyrene derivative, a perylene derivative, a tetracene derivative, an anthracene derivative, and rubrene), a quinacridone derivative, a coumarin derivative, a stilbene derivative, an organic aluminum complex such as tris(8-quinolinolato)aluminum, an iridium complex, a platinum complex, a rhenium complex, a copper complex, a europium complex, a ruthenium complex, and polymer derivatives such as a poly(phenylenevinylene) derivative, a poly(fluorene) derivative, and a poly(phenylene) derivative.

[0062] Detailed examples of compounds used as the light emitting material will be shown below. The material is not limited to these.

##STR00004## ##STR00005## ##STR00006## ##STR00007## ##STR00008## ##STR00009##
##STR00010## ##STR00011## ##STR00012##

[0063] If the light emitting material is a hydrocarbon compound, this is suitable because it is possible to reduce lowering of light emission efficiency caused by exciplex formation or lowering of color purity due to a change of the light emission spectrum of the light emitting material caused by exciplex formation.

[0064] The hydrocarbon compound is a compound made of only carbon and hydrogen, and includes BD7, BD8, GD5 to GD9, and RD1 in the compounds exemplified above.

[0065] If the light emitting material is a condensed polycyclic compound including a 5-membered ring, this is suitable because oxidation hardly occurs because of a high ionization potential, and a long-life element with high durability can be obtained. This includes BD7, BD8, GD5 to GD9, and RD1 in the compounds exemplified above.

[0066] Examples of the light emitting layer host or the light emission assist material contained in the light emitting layer are an aromatic hydrocarbon compound or its derivative, a carbazole derivative, a dibenzofuran derivative, a dibenzothiophene derivative, an organic aluminum complex such as tris(8-quinolinolato)aluminum, and an organic beryllium complex.

[0067] Detailed examples of compounds used as the light emitting layer host or the light emission assist material contained in the light emitting layer will be shown below. The material is not limited to these.

##STR00013## ##STR00014## ##STR00015##

[0068] The host material may be a hydrocarbon compound. The hydrocarbon compound is a compound made of only carbon and hydrogen, and includes EM1 to EM12 and EM16 to EM27 in the compounds exemplified above. As the host material, a material that has, in a single bond that bonds an aryl group unit in its structure, no carbon-heteroatom bonds, like F3 in compound 1, is suitable from the viewpoint of stability.

[0069] The electron transport material can arbitrarily be selected from materials capable of transporting electrons injected from the cathode to the light emitting layer, and is selected in consideration of balance to the hole mobility of the hole transport material. Examples of the material having electron transport performance are an oxadiazole derivative, an oxazole derivative, a pyrazine derivative, a triazole derivative, a triazine derivative, a quinoline derivative, a quinoxaline derivative, a phenanthroline derivative, an organic aluminum complex, and condensed-ring compounds (for example, a fluorene derivative, a naphthalene derivative, a chrysene derivative, and an anthracene derivative). The above-described electron transport material is suitably used for the hole blocking layer as well.

[0070] Detailed examples of compounds used as the electron transport material will be shown below. The material is not limited to these.

##STR00016## ##STR00017##

[0071] The electron injection material can arbitrarily be selected from materials capable of facilitating electron injection from the cathode, and is selected in consideration of balance to hole injection. The organic compound includes an n-type dopant and a reducible dopant. Examples are a compound containing an alkali metal such as lithium fluoride, a lithium complex such as a lithium-quinolinol complex, a benzo-imidazolidene derivative, an imidazolidene derivative, a fulvalene derivative, and an acridine derivative.

[0072] The electron injection material can also be used together with the above-described electron transport material.

Configuration of Organic Light Emitting Element

[0073] The organic light emitting element is provided by forming an insulating layer, a first electrode, an organic compound layer, and a second electrode on a substrate. A protection layer, a color filter, a microlens, and the like may be provided on a cathode. If a color filter is provided, a planarizing layer may be provided between the protection layer and the color filter. The planarizing layer can be formed using acrylic resin or the like. The same applies to a case where a planarizing layer is provided between the color filter and the microlens.

Substrate

[0074] Quartz, glass, a silicon wafer, a resin, a metal, or the like may be used as a substrate. Furthermore, a switching element such as a transistor, a wiring pattern, and the like may be provided on the substrate, and an insulating layer may be provided thereon. The insulating layer may be made of any material as long as a contact hole can be formed so that the wiring pattern can be formed between the first electrode and the substrate and insulation from the unconnected wiring pattern can be ensured. For example, a resin such as polyimide, silicon oxide, silicon nitride, or the like may be used for the insulating layer.

Electrode

[0075] A pair of electrodes can be used as the electrodes. The pair of electrodes can be an anode and a cathode. If an electric field is applied in the direction in which the organic light emitting element emits light, the electrode having a high potential is the anode, and the other is the cathode. It can also be said that the electrode that supplies holes to the light emitting layer is the anode and the electrode that supplies electrons is the cathode.

[0076] As the constituent material of the anode, a material having a large work function may be selected. For example, a metal such as gold, platinum, silver, copper, nickel, palladium, cobalt, selenium, vanadium, or tungsten, a mixture containing some of them, an alloy obtained by combining some of them, or a metal oxide such as tin oxide, zinc oxide, indium oxide, indium tin oxide (ITO), or zinc indium oxide can be used. Furthermore, a conductive polymer such as polyaniline, polypyrrole, or polythiophene can also be used as the constituent material of the anode.

[0077] One of these electrode materials may be used singly, or two or more of them may be used in combination. The anode may be formed by a single layer or a plurality of layers.

[0078] If the electrode is used as a reflective electrode, for example, chromium, aluminum, silver, titanium, tungsten, molybdenum, an alloy thereof, a stacked layer thereof, or the like can be used. The above materials can function as a reflective film having no role as an electrode. If a transparent electrode is used as the electrode, an oxide transparent conductive layer made of indium tin oxide (ITO), indium zinc oxide, or the like can be used, but the present invention is not limited thereto. A photolithography technique can be used to form the electrode.

[0079] On the other hand, as the constituent material of the cathode, a material having a small work function may be selected. Examples of the material include an alkali metal such as lithium, an alkaline earth metal such as calcium, a metal such as aluminum, titanium, manganese, silver, lead, or chromium, and a mixture containing some of them. Alternatively, an alloy obtained by

combining these metals can also be used. For example, a magnesium-silver alloy, an aluminum-lithium alloy, an aluminum-magnesium alloy, a silver-copper alloy, a zinc-silver alloy, or the like can be used. A metal oxide such as indium tin oxide (ITO) can also be used. One of these electrode materials may be used singly, or two or more of them may be used in combination. The cathode may have a single-layer structure or a multilayer structure. Silver may be used as the cathode. To suppress aggregation of silver, a silver alloy may be used. The ratio of the alloy is not limited as long as aggregation of silver can be suppressed. For example, the ratio between silver and another metal may be 1:1, 3:1, or the like.

[0080] The cathode may be a top emission element using an oxide conductive layer made of ITO or the like, or may be a bottom emission element using a reflective electrode made of aluminum (Al) or the like, and is not particularly limited. The method of forming the cathode is not particularly limited, but if direct current sputtering or alternating current sputtering is used, the good coverage is achieved for the film to be formed, and the resistance of the cathode can be lowered.

Pixel Isolation Layer

[0081] A pixel isolation layer may be formed by a so-called silicon oxide, such as silicon nitride (SiN), silicon oxynitride (SiON), or silicon oxide (SiO), formed using a Chemical Vapor Deposition (CVD) method. To increase the resistance in the in-plane direction of the organic compound layer, the organic compound layer, especially the hole transport layer may be thinly deposited on the side wall of the pixel isolation layer. More specifically, the organic compound layer can be deposited so as to have a thin film thickness on the side wall by increasing the taper angle of the side wall of the pixel isolation layer or the film thickness of the pixel isolation layer to increase vignetting during vapor deposition.

[0082] On the other hand, the taper angle of the side wall of the pixel isolation layer or the film thickness of the pixel isolation layer can be adjusted to the extent that no space is formed in the protection layer formed on the pixel isolation layer. Since no space is formed in the protection layer, it is possible to reduce generation of defects in the protection layer. Since generation of defects in the protection layer is reduced, a decrease in reliability caused by generation of a dark spot or occurrence of a conductive failure of the second electrode can be reduced.

[0083] According to this embodiment, even if the taper angle of the side wall of the pixel isolation layer is not acute, it is possible to effectively suppress leakage of charges to an adjacent pixel. As a result of this consideration, it has been found that the taper angle of 60° (inclusive) to 90° (inclusive) can sufficiently reduce the occurrence of defects. The film thickness of the pixel isolation layer may be 10 nm (inclusive) to 150 nm (inclusive). A similar effect can be obtained in a configuration including only pixel electrodes without the pixel isolation layer. However, in this case, the film thickness of the pixel electrode is set to be equal to or smaller than half the film thickness of the organic layer or the end portion of the pixel electrode is formed to have a forward tapered shape of less than 60°. With this, short circuit of the organic light emitting element can be reduced.

[0084] Furthermore, in a case where the first electrode is the cathode and the second electrode is the anode, a high color gamut and low-voltage driving can be achieved by forming the electron transport material and charge transport layer and forming the light emitting layer on the charge transport layer.

Organic Compound Layer

[0085] The organic compound layer may be formed by a single layer or a plurality of layers. If the organic compound layer includes a plurality of layers, the layers can be called a hole injection layer, a hole transport layer, an electron blocking layer, a light emitting layer, a hole blocking layer, an electron transport layer, and an electron injection layer in accordance with the functions of the layers. The organic compound layer is mainly formed from an organic compound but may contain inorganic atoms and an inorganic compound. For example, the organic compound layer may contain copper, lithium, magnesium, aluminum, iridium, platinum, molybdenum, zinc, or the like.

The organic compound layer may be arranged between the first and second electrodes, and may be arranged in contact with the first and second electrodes. If a plurality of light emitting layers are provided, a charge generation portion may be arranged between the first light emitting layer and the second light emitting layer. The charge generation portion may contain an organic compound with a lowest unoccupied molecular orbital energy (LUMO) of -5.0 eV or less. The same applies to a case where a charge generating portion is provided between the second light emitting layer and the third light emitting layer.

Protection Layer

[0086] A protection layer may be provided on the cathode. For example, by adhering glass provided with a moisture absorbing agent on the cathode, permeation of water or the like into the organic compound layer can be suppressed and occurrence of display defects can be suppressed. Furthermore, as another embodiment, a passivation layer made of silicon nitride or the like may be provided on the cathode to suppress permeation of water or the like into the organic compound layer. For example, the protection layer can be formed by forming the cathode, transferring it to another chamber without breaking the vacuum, and forming silicon nitride having a thickness of $2\text{ }\mu\text{m}$ by the CVD method. The protection layer may be provided using an atomic layer deposition (ALD) method after deposition of the protection layer using the CVD method. The material of the protection layer by the ALD method is not limited but can be silicon nitride, silicon oxide, aluminum oxide, or the like. Silicon nitride may further be formed by the CVD method on the protection layer formed by the ALD method. The protection layer formed by the ALD method may have a film thickness smaller than that of the protection layer formed by the CVD method. More specifically, the film thickness of the protection layer formed by the ALD method may be 50% or less, or 10% or less of that of the protection layer formed by the CVD method.

Color Filter

[0087] A color filter may be provided on the protection layer. For example, a color filter considering the size of the organic light emitting element may be provided on another substrate, and the substrate with the color filter formed thereon may be bonded to the substrate with the organic light emitting element provided thereon. Alternatively, for example, a color filter may be patterned on the above-described protection layer using a photolithography technique. The color filter may be formed from a polymeric material.

Planarizing Layer

[0088] A planarizing layer may be arranged between the color filter and the protection layer. The planarizing layer is provided to reduce unevenness of the layer below the planarizing layer. The planarizing layer may be called a material resin layer without limiting the purpose of the layer. The planarizing layer may be formed from an organic compound, and may be made of a low-molecular material or a polymeric material. In consideration of reduction of unevenness, a polymeric organic compound may be used for the planarizing layer.

[0089] The planarizing layers may be provided above and below the color filter. In that case, the same or different constituent materials may be used for these planarizing layers. More specifically, examples of the material of the planarizing layer include polyvinyl carbazole resin, polycarbonate resin, polyester resin, ABS resin, acrylic resin, polyimide resin, phenol resin, epoxy resin, silicone resin, and urea resin.

Microlens

[0090] The organic light emitting device may include an optical member such as a microlens on the light emission side. The microlens can be made of acrylic resin, epoxy resin, or the like. The microlens can aim to increase the amount of light extracted from the organic light emitting device and control the direction of light to be extracted. The microlens can have a hemispherical shape. If the microlens has a hemispherical shape, among tangents contacting the hemisphere, there is a tangent parallel to the insulating layer, and the contact between the tangent and the hemisphere is the vertex of the microlens. The vertex of the microlens can be decided in the same manner even in

an arbitrary sectional view. That is, among tangents contacting the semicircle of the microlens in a sectional view, there is a tangent parallel to the insulating layer, and the contact between the tangent and the semicircle is the vertex of the microlens.

[0091] Furthermore, the middle point of the microlens can also be defined. In the section of the microlens, a line segment from a point at which an arc shape ends to a point at which another arc shape ends is assumed, and the middle point of the line segment can be called the middle point of the microlens. A section for determining the vertex and the middle point may be a section perpendicular to the insulating layer.

[0092] The microlens includes a first surface including a convex portion and a second surface opposite to the first surface. The second surface can be arranged on the functional layer (light emitting layer) side of the first surface. For this configuration, the microlens needs to be formed on the light emitting device. If the functional layer is an organic layer, a process which produces high temperature in the manufacturing step of the microlens may be avoided. In addition, if it is configured to arrange the second surface on the functional layer side of the first surface, all the glass transition temperatures of an organic compound forming the organic layer may be 100° C. or more. For example, 130° C. or more is suitable.

Counter Substrate

[0093] A counter substrate may be arranged on the planarizing layer. The counter substrate is called a counter substrate because it is provided at a position corresponding to the above-described substrate. The constituent material of the counter substrate can be the same as that of the above-described substrate. If the above-described substrate is the first substrate, the counter substrate can be the second substrate.

Organic Layer

[0094] The organic compound layer (hole injection layer, hole transport layer, electron blocking layer, light emitting layer, hole blocking layer, electron transport layer, electron injection layer, and the like) forming the organic light emitting element according to an embodiment of the present disclosure may be formed by the method to be described below.

[0095] The organic compound layer forming the organic light emitting element according to the embodiment of the present disclosure can be formed by a dry process using a vacuum deposition method, an ionization deposition method, a sputtering method, a plasma method, or the like. Instead of the dry process, a wet process that forms a layer by dissolving a solute in an appropriate solvent and using a well-known coating method (for example, a spin coating method, a dipping method, a casting method, an LB method, an inkjet method, or the like) can be used.

[0096] Here, when the layer is formed by a vacuum deposition method, a solution coating method, or the like, crystallization or the like hardly occurs and excellent temporal stability is obtained. Furthermore, when the layer is formed using a coating method, it is possible to form the film in combination with a suitable binder resin.

[0097] Examples of the binder resin include polyvinyl carbazole resin, polycarbonate resin, polyester resin, ABS resin, acrylic resin, polyimide resin, phenol resin, epoxy resin, silicone resin, and urea resin. However, the binder resin is not limited to them.

[0098] One of these binder resins may be used singly as a homopolymer or a copolymer, or two or more of them may be used in combination. Furthermore, additives such as a well-known plasticizer, antioxidant, and an ultraviolet absorber may also be used as needed.

Pixel Circuit

[0099] The light emitting device can include a pixel circuit connected to the light emitting element. The pixel circuit may be an active matrix circuit that individually controls light emission of the first and second light emitting elements. The active matrix circuit may be a voltage or current programming circuit. A driving circuit includes a pixel circuit for each pixel. The pixel circuit can include a light emitting element, a transistor for controlling light emission luminance of the light emitting element, a transistor for controlling a light emission timing, a capacitor for holding the

gate voltage of the transistor for controlling the light emission luminance, and a transistor for connection to GND without intervention of the light emitting element.

[0100] The light emitting device includes a display region and a peripheral region arranged around the display region. The light emitting device includes the pixel circuit in the display region and a display control circuit in the peripheral region. The mobility of the transistor forming the pixel circuit may be smaller than that of a transistor forming the display control circuit.

[0101] The slope of the current-voltage characteristic of the transistor forming the pixel circuit may be smaller than that of the current-voltage characteristic of the transistor forming the display control circuit. The slope of the current-voltage characteristic can be measured by a so-called V_g - I_g characteristic.

[0102] The transistor forming the pixel circuit is a transistor connected to the light emitting element such as the first light emitting element.

Pixel

[0103] The organic light emitting device includes a plurality of pixels. Each pixel includes sub-pixels that emit light components of different colors. The sub-pixels may include, for example, R, G, and B emission colors, respectively.

[0104] In each pixel, a region also called a pixel opening emits light. The pixel opening can have a size of 5 μm (inclusive) to 15 μm (inclusive). More specifically, the pixel opening can have a size of 11 μm , 9.5 μm , 7.4 μm , 6.4 μm , or the like.

[0105] A distance between the sub-pixels can be 10 μm or less, and can be, more specifically, 8 μm , 7.4 μm , or 6.4 μm .

[0106] The pixels can have a known arrangement form in a plan view. For example, the pixels may have a stripe arrangement, a delta arrangement, a pentile arrangement, or a Bayer arrangement. The shape of each sub-pixel in a plan view may be any known shape. For example, a quadrangle such as a rectangle or a rhombus, a hexagon, or the like may be possible. A shape which is not a correct shape but is close to a rectangle is included in a rectangle, as a matter of course. The shape of the sub-pixel and the pixel arrangement can be used in combination.

Application of Organic Light Emitting Element of Embodiment of Present Disclosure

[0107] The organic light emitting element according to an embodiment of the present disclosure can be used as a constituent member of a display device or an illumination device. In addition, the organic light emitting element is applicable to the exposure light source of an electrophotographic image forming device, the backlight of a liquid crystal display device, a light emitting device including a color filter in a white light source, and the like.

[0108] The display device may be an image information processing device that includes an image input unit for inputting image information from an area CCD, a linear CCD, a memory card, or the like, and an information processing unit for processing the input information, and displays the input image on a display unit.

[0109] In addition, a display unit included in an image capturing device or an inkjet printer can have a touch panel function. The driving type of the touch panel function may be an infrared type, a capacitance type, a resistive film type, or an electromagnetic induction type, and is not particularly limited. The display device may be used for the display unit of a multifunction printer.

[0110] More details will be described next with reference to the accompanying drawings. FIG. 9A shows an example of the pixel PIX arranged in the light emitting device 100. The pixel includes sub-pixels 810 (pixels PIX). The sub-pixels are divided into sub-pixels 810R, 810G, and 810B by emitted light components. The light emission colors may be discriminated by the wavelengths of light components emitted from the light emitting layers, or light emitted from each sub-pixel may be selectively transmitted or undergo color conversion by a color filter or the like. Each sub-pixel includes a reflective electrode 802 as the first electrode on an interlayer insulating layer 801, an insulating layer 803 covering the end of the reflective electrode 802, an organic compound layer 804 covering the first electrode and the insulating layer, a transparent electrode 805 as the second

electrode, a protection layer **806**, and a color filter **807**.

[0111] The interlayer insulating layer **801** can include a transistor and a capacitive element arranged in the interlayer insulating layer **801** or a layer below it. The transistor and the first electrode can electrically be connected via a contact hole (not shown) or the like.

[0112] The insulating layer **803** can also be called a bank or a pixel isolation film. The insulating layer **803** covers the end of the first electrode, and is arranged to surround the first electrode. A portion of the first electrode where no insulating layer **803** is arranged is in contact with the organic compound layer **804** to form a light emitting region.

[0113] The organic compound layer **804** includes a hole injection layer **841**, a hole transport layer **842**, a first light emitting layer **843**, a second light emitting layer **844**, and an electron transport layer **845**.

[0114] The second electrode may be a transparent electrode, a reflective electrode, or a semi-transmissive electrode.

[0115] The protection layer **806** suppresses permeation of water into the organic compound layer. The protection layer is shown as a single layer but may include a plurality of layers. Each layer can be an inorganic compound layer or an organic compound layer.

[0116] The color filter **807** is divided into color filters **807R**, **807G**, and **807B** by colors. The color filters can be formed on a planarizing film (not shown). A resin protection layer (not shown) may be arranged on the color filters. The color filters can be formed on the protection layer **806**.

Alternatively, the color filters can be provided on the counter substrate such as a glass substrate, and then the substrate may be bonded.

[0117] The display device **800** (corresponding to the above-described light emitting device **100**) shown in FIG. **9B** is provided with an organic light emitting element **826** as an example of a light emitting element and a TFT **818** as an example of a transistor. A substrate **811** of glass, silicon, or the like is provided and an insulating layer **812** is provided on the substrate **811**. The active element such as the TFT **818** is arranged on the insulating layer, and a gate electrode **813**, a gate insulating film **814**, and a semiconductor layer **815** of the active element are arranged. The TFT **818** further includes the semiconductor layer **815**, a drain electrode **816**, and a source electrode **817**. An insulating film **819** is provided on the TFT **818**. The source electrode **817** and an anode **821** forming the organic light emitting element **826** are connected via a contact hole **820** formed in the insulating film.

[0118] A method of electrically connecting the electrodes (anode and cathode) included in the organic light emitting element **826** and the electrodes (source electrode and drain electrode) included in the TFT is not limited to that shown in FIG. **9B**. That is, one of the anode and cathode and one of the source electrode and drain electrode of the TFT are electrically connected. The TFT indicates a thin-film transistor.

[0119] In the display device **800** shown in FIG. **9B**, an organic compound layer is illustrated as one layer. However, an organic compound layer **822** may include a plurality of layers. A first protection layer **824** and a second protection layer **825** are provided on a cathode **823** to suppress deterioration of the organic light emitting element.

[0120] A transistor is used as a switching element in the display device **800** shown in FIG. **9B**, but another switching element may be used instead.

[0121] The transistor used in the display device **800** shown in FIG. **9B** is not limited to a transistor using a single-crystal silicon wafer, and may be a thin-film transistor including an active layer on an insulating surface of a substrate. Examples of the active layer include single-crystal silicon, amorphous silicon, non-single-crystal silicon such as microcrystalline silicon, and a non-single-crystal oxide semiconductor such as indium zinc oxide and indium gallium zinc oxide. Note that a thin-film transistor is also called a TFT element.

[0122] The transistor included in the display device **800** shown in FIG. **9B** may be formed in the substrate such as a silicon substrate. Forming the transistor in the substrate means forming the

transistor by processing the substrate such as a silicon substrate. That is, when the transistor is included in the substrate, it can be considered that the substrate and the transistor are formed integrally.

[0123] The light emission luminance of the organic light emitting element according to this embodiment can be controlled by the TFT which is an example of a switching element, and the plurality of organic light emitting elements can be provided in a plane to display an image with the light emission luminances of the respective elements. Here, the switching element according to this embodiment is not limited to the TFT, and may be a transistor formed from low-temperature polysilicon or an active matrix driver formed on the substrate such as a silicon substrate. The term “on the substrate” may mean “in the substrate”. Whether to provide a transistor in the substrate or use a TFT is selected based on the size of the display unit. For example, if the size is about 0.5 inch, the organic light emitting element may be provided on the silicon substrate.

[0124] FIG. **10** is a schematic view showing an example of the display device using the light emitting device **100** according to this embodiment. A display device **1000** can include a touch panel **1003**, a display panel **1005**, a frame **1006**, a circuit board **1007**, and a battery **1008** between an upper cover **1001** and a lower cover **1009**. Flexible printed circuits (FPCs) **1002** and **1004** are respectively connected to the touch panel **1003** and the display panel **1005**. Active elements such as transistors are arranged on the circuit board **1007**. The battery **1008** is unnecessary if the display device **1000** is not a portable apparatus. Even when the display device **1000** is a portable apparatus, the battery **1008** need not be provided at this position. The light emitting device **100** can be applied to the display panel **1005**. The pixels PIX arranged in the light emitting device **100** functioning as the display panel **1005** are connected to a control circuit including the active elements such as transistors arranged on the circuit board **1007** and operate.

[0125] The display device **1000** shown in FIG. **10** can be used for a display unit of a photoelectric conversion device (also referred to as an image capturing device) including an optical unit having a plurality of lenses, and an image sensor for receiving light having passed through the optical unit and photoelectrically converting the light into an electric signal. The photoelectric conversion device can include a display unit for displaying information acquired by the image sensor. In addition, the display unit can be either a display unit exposed outside the photoelectric conversion device, or a display unit arranged in the finder. The photoelectric conversion device can be a digital camera or a digital video camera.

[0126] FIG. **11** is a schematic view showing an example of the photoelectric conversion device using the light emitting device **100** according to this embodiment. A photoelectric conversion device **1100** can include a viewfinder **1101**, a rear display **1102**, an operation unit **1103**, and a housing **1104**. The photoelectric conversion device **1100** can also be called an image capturing device. The light emitting device **100** according to this embodiment can be applied to the viewfinder **1101** or the rear display **1102** as a display unit. In this case, the light emitting device **100** can display not only an image to be captured but also environment information, image capturing instructions, and the like. Examples of the environment information are the intensity and direction of external light, the moving velocity of an object, and the possibility that an object is covered with an obstacle.

[0127] Since the timing suitable for image capturing is a very short time in many cases, it is better to display the information as soon as possible. Therefore, the light emitting device **100** in which the pixel PIX including the light emitting element using the organic light emitting material such as an organic EL element is arranged may be used for the viewfinder **1101** or the rear display **1102**. This is so because the organic light emitting material has a high response speed. The light emitting device **100** using the organic light emitting material can be used for the devices that require a high display speed more suitably than for the liquid crystal display device.

[0128] The photoelectric conversion device **1100** includes an optical unit (not shown). This optical unit has a plurality of lenses, and forms an image on a photoelectric conversion element (not

shown) that receives light having passed through the optical unit and is accommodated in the housing **1104**. The focal points of the plurality of lenses can be adjusted by adjusting the relative positions. This operation can also automatically be performed.

[0129] The light emitting device **100** may be applied to a display unit of an electronic apparatus. At this time, the display unit can have both a display function and an operation function. Examples of the portable terminal are a portable phone such as a smartphone, a tablet, and a head mounted display.

[0130] FIG. **12** is a schematic view showing an example of an electronic apparatus using the light emitting device **100** according to this embodiment. An electronic apparatus **1200** includes a display unit **1201**, an operation unit **1202**, and a housing **1203**. The housing **1203** can accommodate a circuit, a printed board having this circuit, a battery, and a communication unit. The operation unit **1202** can be a button or a touch-panel-type reaction unit. The operation unit **1202** can also be a biometric authentication unit that performs unlocking or the like by authenticating the fingerprint. The portable apparatus including the communication unit can also be regarded as a communication apparatus. The light emitting device **100** according to this embodiment can be applied to the display unit **1201**.

[0131] FIGS. **13A** and **13B** are schematic views showing examples of the display device using the light emitting device **100** according to this embodiment. FIG. **13A** shows a display device such as a television monitor or a PC monitor. A display device **1300** includes a frame **1301** and a display unit **1302**. The light emitting device **100** according to this embodiment can be applied to the display unit **1302**. The display device **1300** can include a base **1303** that supports the frame **1301** and the display unit **1302**. The base **1303** is not limited to the form shown in FIG. **13A**. For example, the lower side of the frame **1301** may also function as the base **1303**. In addition, the frame **1301** and the display unit **1302** can be bent. The radius of curvature in this case can be 5,000 mm (inclusive) to 6,000 mm (inclusive).

[0132] FIG. **13B** is a schematic view showing another example of the display device using the light emitting device **100** according to this embodiment. A display device **1310** shown in FIG. **13B** can be folded, and is a so-called foldable display device. The display device **1310** includes a first display unit **1311**, a second display unit **1312**, a housing **1313**, and a bending point **1314**. The light emitting device **100** according to this embodiment can be applied to each of the first display unit **1311** and the second display unit **1312**. The first display unit **1311** and the second display unit **1312** can also be one seamless display device. The first display unit **1311** and the second display unit **1312** can be divided by the bending point. The first display unit **1311** and the second display unit **1312** can display different images, and can also display one image together.

[0133] FIG. **14** is a schematic view showing an example of the illumination device using the light emitting device **100** according to this embodiment. An illumination device **1400** can include a housing **1401**, a light source **1402**, a circuit board **1403**, an optical film **1404**, and a light diffusing unit **1405**. The light emitting device **100** according to this embodiment can be applied to the light source **1402**. The optical film **1404** can be a filter that improves the color rendering of the light source. When performing lighting-up or the like, the light diffusing unit **1405** can throw the light of the light source over a broad range by effectively diffusing the light. The illumination device can also include a cover on the outermost portion, as needed. The illumination device **1400** can include both or one of the optical film **1404** and the light diffusing unit **1405**.

[0134] The illumination device **1400** is, for example, a device for illuminating the interior of the room. The illumination device **1400** can emit white light, natural white light, or light of any color from blue to red. The illumination device **1400** can also include a light control circuit for controlling these light components. The illumination device **1400** can also include a power supply circuit connected to the light emitting device **100** functioning as the light source **1402**. The power supply circuit is a circuit for converting an AC voltage into a DC voltage. White has a color temperature of 4,200 K, and natural white has a color temperature of 5,000 K. The illumination

device **1400** may also include a color filter. In addition, the illumination device **1400** can include a heat radiation unit. The heat radiation unit radiates the internal heat of the device to the outside of the device, and examples are a metal having a high specific heat and liquid silicon.

[0135] FIG. **15** is a schematic view of an automobile having a taillight as an example of a vehicle lighting appliance using the light emitting device **100** according to this embodiment. An automobile **1500** has a taillight **1501**, and can have a form in which the taillight **1501** is turned on when performing a braking operation or the like. The light emitting device **100** according to this embodiment can be used as a headlight serving as a vehicle lighting appliance. The automobile is an example of a moving body, and the moving body may be a ship, a drone, an aircraft, a railroad car, an industrial robot, or the like. The moving body may include a main body and a lighting appliance provided in the main body. The lighting appliance may be used to make a notification of the current position of the main body.

[0136] The light emitting device **100** according to this embodiment can be applied to the taillight **1501**. The taillight **1501** can include a protection member for protecting the light emitting device **100** functioning as the taillight **1501**. The material of the protection member is not limited as long as the material is a transparent material with a strength that is somewhat high, and an example is polycarbonate. The protection member may be made of a material obtained by mixing a furandicarboxylic acid derivative, an acrylonitrile derivative, or the like in polycarbonate.

[0137] The automobile **1500** can include a vehicle body **1503**, and a window **1502** attached to the vehicle body **1503**. This window can be a window for checking the front and back of the automobile, and can also be a transparent display such as a head-up display. For this transparent display, the light emitting device **100** according to this embodiment may be used. In this case, the constituent materials of the electrodes and the like of the light emitting device **100** are formed by transparent members.

[0138] Further application examples of the light emitting device **100** according to this embodiment will be described with reference to FIGS. **16A** and **16B**. The light emitting device **100** can be applied to a system that can be worn as a wearable device such as smartglasses, a Head Mounted Display (HMD), or a smart contact lens. An image capturing display device used for such application examples includes an image capturing device capable of photoelectrically converting visible light and a light emitting device capable of emitting visible light.

[0139] Glasses **1600** (smartglasses) according to one application example will be described with reference to FIG. **16A**. An image capturing device **1602** such as a CMOS sensor or an SPAD is provided on the surface side of a lens **1601** of the glasses **1600**. In addition, the light emitting device **100** according to this embodiment is provided on the back surface side of the lens **1601**.

[0140] The glasses **1600** further include a control device **1603**. The control device **1603** functions as a power supply that supplies electric power to the image capturing device **1602** and the light emitting device **100** according to each embodiment. In addition, the control device **1603** controls the operations of the image capturing device **1602** and the light emitting device **100**. An optical system configured to condense light to the image capturing device **1602** is formed on the lens **1601**.

[0141] Glasses **1610** (smartglasses) according to one application example will be described with reference to FIG. **16B**. The glasses **1610** include a control device **1612**, and an image capturing device corresponding to the image capturing device **1602** and the light emitting device **100** are mounted on the control device **1612**. The image capturing device in the control device **1612** and an optical system configured to project light emitted from the light emitting device **100** are formed in a lens **1611**, and an image is projected to the lens **1611**. The control device **1612** functions as a power supply that supplies electric power to the image capturing device and the light emitting device **100**, and controls the operations of the image capturing device and the light emitting device **100**. The control device **1612** may include a line-of-sight detection unit that detects the line of sight of a wearer. The detection of a line of sight may be done using infrared rays. An infrared ray emitting unit emits infrared rays to an eyeball of the user who is gazing at a displayed image. An

image capturing unit including a light receiving element detects reflected light of the emitted infrared rays from the eyeball, thereby obtaining a captured image of the eyeball. A reduction unit for reducing light from the infrared ray emitting unit to the display unit in a planar view is provided, thereby reducing deterioration of image quality.

[0142] The line of sight of the user to the displayed image is detected from the captured image of the eyeball obtained by capturing the infrared rays. An arbitrary known method can be applied to the line-of-sight detection using the captured image of the eyeball. As an example, a line-of-sight detection method based on a Purkinje image obtained by reflection of irradiation light by a cornea can be used.

[0143] More specifically, line-of-sight detection processing based on pupil center corneal reflection is performed. Using pupil center corneal reflection, a line-of-sight vector representing the direction (rotation angle) of the eyeball is calculated based on the image of the pupil and the Purkinje image included in the captured image of the eyeball, thereby detecting the line-of-sight of the user.

[0144] The light emitting device **100** according to the embodiment of the present disclosure can include an image capturing device including a light receiving element, and control a displayed image based on the line-of-sight information of the user from the image capturing device.

[0145] More specifically, the light emitting device **100** decides a first visual field region at which the user is gazing and a second visual field region other than the first visual field region based on the line-of-sight information. The first visual field region and the second visual field region may be decided by the control device of the light emitting device **100**, or those decided by an external control device may be received. In the display region of the light emitting device **100**, the display resolution of the first visual field region may be controlled to be higher than the display resolution of the second visual field region. That is, the resolution of the second visual field region may be lower than that of the first visual field region.

[0146] In addition, the display region includes a first display region and a second display region different from the first display region, and a region of higher priority is decided from the first display region and the second display region based on line-of-sight information. The first display region and the second display region may be decided by the control device of the light emitting device **100**, or those decided by an external control device may be received. The resolution of the region of higher priority may be controlled to be higher than the resolution of the region other than the region of higher priority. That is, the resolution of the region of relatively low priority may be low.

[0147] Note that AI may be used to decide the first visual field region or the region of higher priority. The AI may be a model configured to estimate the angle of the line of sight and the distance to a target ahead the line of sight from the image of the eyeball using the image of the eyeball and the direction of actual viewing of the eyeball in the image as supervised data. The AI program may be held by the light emitting device **100**, the image capturing device, or an external device. If the external device holds the AI program, it is transmitted to the light emitting device **100** via communication.

[0148] When performing display control based on line-of-sight detection, smartglasses further including an image capturing device configured to capture the outside can be applied. The smartglasses can display captured outside information in real time.

[0149] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0150] This application claims the benefit of Japanese Patent Application No. 2024-025574, filed Feb. 22, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A light emitting device comprising, on a main surface of a semiconductor substrate, a display region in which a plurality of pixels are arranged, wherein the main surface includes a first region in which a first conductive pattern is arranged, and a second region which is adjacent to the first region and in which a second conductive pattern is arranged, each of the plurality of pixels includes an organic light emitting element, and a driving transistor configured to supply a current according to a luminance signal to the organic light emitting element, each of the first conductive pattern and the second conductive pattern includes a conductive pattern connected to a gate electrode of the driving transistor in one of the plurality of pixels, the first region and the second region are adjacent at a boundary with a deviation between the first conductive pattern and the second conductive pattern, and the boundary does not pass through a center of the display region.
2. The device according to claim 1, wherein each of the first conductive pattern and the second conductive pattern includes a pattern arranged in a wiring layer closest to the gate electrode among a plurality of wiring layers arranged in the display region.
3. The device according to claim 1, wherein each of the first conductive pattern and the second conductive pattern includes a pattern arranged in a second closest wiring layer to the gate electrode among a plurality of wiring layers arranged in the display region.
4. The device according to claim 1, wherein each of the first conductive pattern and the second conductive pattern includes at least one of a plug connected to the gate electrode, a plug connected to a pattern arranged in a wiring layer closest to the gate electrode among a plurality of wiring layers arranged in the display region, and a plug connected to a pattern arranged in a second closest wiring layer to the gate electrode among the plurality of wiring layers.
5. The device according to claim 1, wherein each of the plurality of pixels further includes a write transistor configured to write the luminance signal to the gate electrode, and each of the first conductive pattern and the second conductive pattern includes a conductive pattern connected to a gate electrode of the write transistor in one of the plurality of pixels.
6. The device according to claim 1, wherein the boundary passes through the display region.
7. The device according to claim 1, wherein the boundary does not pass through the display region.
8. The device according to claim 1, wherein in an orthogonal projection to the main surface, an outer edge of the semiconductor substrate includes a first portion, in a direction crossing a direction in which the boundary extends, the boundary is arranged between the first portion and the center, and a length between the boundary and the center is not less than a length between the boundary and the first portion.
9. The device according to claim 1, wherein in an orthogonal projection to the main surface, the display region has a rectangular shape, in a direction crossing a direction in which the boundary extends, the boundary is arranged between the center and a first side among sides that form an outer edge of the display region, and a length between the boundary and the center is not less than a length between the boundary and the first side.
10. The device according to claim 1, further comprising an eyepiece optical system, wherein in an orthogonal projection to the main surface, the display region has a rectangular shape, in a direction crossing a direction in which the boundary extends, the boundary is arranged between the center and a first side among sides that form an outer edge of the display region, and if a visual field angle through the eyepiece optical system is defined as A degree, the boundary is arranged within a range of $\{(A/2-30)/A\} \times 100[\%]$ from the first side with respect to a length of the display region in the crossing direction.
11. The device according to claim 1, further comprising an eyepiece optical system, wherein in an orthogonal projection to the main surface, the display region has a rectangular shape, in a direction crossing a direction in which the boundary extends, the boundary is arranged between the center

and a first side among sides that form an outer edge of the display region, and the boundary is arranged within a range of not less than 30° in the crossing direction from a center of a visual field angle through the eyepiece optical system.

12. The device according to claim 9, wherein the rectangular shape has a long side and a short side, and the first side forms the long side.

13. The device according to claim 9, wherein the rectangular shape has a long side and a short side, and the first side forms the short side.

14. The device according to claim 9, wherein the boundary extends parallel to a direction in which the first side extends.

15. The device according to claim 1, wherein the display region includes a first display region, and a second display region in which a resolution of a displayed image is lower than that of the first display region, and the boundary passes through the second display region.

16. The device according to claim 15, wherein the boundary does not pass through the first display region.

17. The device according to claim 15, wherein an arrangement density of the pixels in the first display region is higher than that in the second display region.

18. A wearable device including a display device configured to display an image, wherein the display device includes the light emitting device according to claim 1.

19. The device according to claim 18, wherein the display device includes a plurality of light emitting devices including a first light emitting device arranged in correspondence with a left eye, and a second light emitting device arranged in correspondence with a right eye when a user wears the wearable device.

20. The device according to claim 19, wherein when the user wears the wearable device, a direction in which the boundary of the first light emitting device is arranged with respect to the center of the display region of the first light emitting device and a direction in which the boundary of the second light emitting device is arranged with respect to the center of the display region of the second light emitting device are different from each other.

21. The device according to claim 19, wherein when the user wears the wearable device, the boundary of the first light emitting device is arranged on a left side with respect to the center of the display region of the first light emitting device, and the boundary of the second light emitting device is arranged on a right side with respect to the center of the display region of the second light emitting device.

22. The device according to claim 19, wherein when the user wears the wearable device, the boundary of each of the first light emitting device and the second light emitting device is arranged on an upper side with respect to the center of the display region of each of the first light emitting device and the second light emitting device.

23. A display device comprising the light emitting device according to claim 1, and a control circuit connected to the light emitting device.

24. A photoelectric conversion device comprising an optical unit including a plurality of lenses, an image sensor configured to receive light having passed through the optical unit, and a display configured to display an image, wherein the display includes the light emitting device according to claim 1.

25. An electronic apparatus comprising a housing provided with a display, and a communication unit provided in the housing and configured to perform external communication, wherein the display includes the light emitting device according to claim 1.

26. An illumination device comprising a light source, and at least one of a light diffusing unit and an optical film, wherein the light source includes the light emitting device according to claim 1.

27. A moving body comprising a main body, and a lighting appliance provided in the main body, wherein the lighting appliance includes the light emitting device according to claim 1.
